Dealer Spreads in the Corporate Bond Market: Agent vs. Market-Making Roles

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Background and Motivation

- ► 90%+ of financial securities are traded in OTC markets.
- ► Intermediaries in OTC markets are ‘broker-dealers’.
  - – As ‘brokers’, they serve as agents for customers: Provide access to market and counterparty search services.
  - – As ‘dealers’, they do ‘market-making’: Buy (or sell) on their own account as principals when a customer wants to sell (or buy).
- ► In a particular trade, a broker-dealer (typically just “dealer”) can function purely as an agent, purely as a ‘market-maker’, or both; and this role can change from trade to trade.
- ► Hence, dealer spreads reflect not just the costs arising from their market-making role, but also the costs arising from their role as agent.

Background and Motivation

- Dealer spreads accordingly reflect:
  - Costs and revenues arising from their market-making role.
    - Adverse information risk (Glosten and Harris, JFE 1988).
    - Inventory management risk (Ho and Stoll, JF 1983).
  - Costs and benefits arising from their role as agent.
    - Counterparty search costs (Duffie, 2012).
    - Direct customer management costs.
    - Information-related benefits arising from interfacing with customers (Naik, Neuberger and Viswanathan, RFS 1999).

Background and Motivation

- The extensive earlier literature on OTC market dealer spreads treated the spreads essentially as just representing market making costs, and tested hypotheses accordingly.
- These studies have:
  - **Not** considered and estimated the size and nature of agent-role related dealers' costs.
  - **Not** tested any hypotheses on agent-related dealer spreads.
  - Have **all** underestimated average overall trading costs by conflating the spreads of dealers acting purely as agents with full spreads that include both agent and market making costs.

Earlier Literature on Dealer Spreads

- US Corporate bond markets: Bessembinder, Maxwell and Venkataraman (JFE 2006); Goldstein, Hotchkiss and Sirri (RFS 2007); Edwards, Harris, and Piwowar (JF 2007) (EHP (2007)).
- US municipal bond market: Harris and Piwowar (JF 2006)
- London government bond market: Naik and Yadav (JF 2003).
- FX markets: Bessembinder (1994) and Lyons (1995).
- NASDAQ: Huang and Stoll (JFE 1996), Barclay, Christie, Harris and Schultz (JF 1999), and Huang (JF 2002).
- London Stock Exchange: Hansch, Naik and Viswanathan (JF 1998, JF 1999) and Naik and Yadav (JFE 2003).

- Nothing on the dealer's role as agent.

In This Paper ….

- We decompose dealer spreads in U.S. corporate bond OTC markets into components arising from:
  - Dealers' market-making role, and
  - Dealers' role as agents for their non-dealer customers.

- …. And, we investigate the determinants of both market-making and agent-related components separately.

In This Paper ….

- We do this by utilizing the large subset of corporate bond market trades that are (what the industry calls) "***Riskless Principal Trades***" – in which dealers act purely as agents.
- An estimated 39.3% of customer buys and 34.2% of customer sells in our two-year sample period of US corporate bonds are "riskless principal trades."
  - Harris (2015) in his paper based on Interactive Brokers transactions has very similar proportions.
- This is clearly a very large fraction of overall trades.

Identifying Riskless Principal Trades

- Suppose dealer C receives a customer sell order for a trade in which it wants to act purely as agent: i.e., does not wish to keep in inventory. C arranges to:
  - buy the bonds from the customer, and,
  - *simultaneously* resell to another dealer B.
  - This is a riskless principal trade for Dealer C.
- Likewise, suppose dealer A receives a customer buy order for a bond that it does not have in inventory, or wish to carry a negative inventory exposure for. A arranges to:
  - buy the bonds from another dealer B, and,
  - *simultaneously* resell the bonds to the customer.
  - This is a riskless principal trade for Dealer A.

Identifying Riskless Principal Trades

- In these riskless principal trades, dealers A and C do not carry any price or inventory risk, and serve purely as agents for external (non-dealer) customers: searching for the best counterparty and managing customer relationships.
- Dealer B serves purely as a market maker, bearing all inventory risks, but not facing any customer interface costs, or search costs other than those related to managing her inventory.
- ***We identify riskless principal trades by identifying such paired-trades in TRACE data on corporate bonds.***

Paired Trades

- Existence of paired trades was first identified by Zitzewitz (WP, 2011), but without any conceptual linkage with agent-related component of the spread.
- We linked paired trades to agent-dealer spreads conceptually and empirically in our 2013 WP.
- Since then, paired trades have been utilized from different perspectives by Harris (2015); Ruzza (2017), Schultz (2017), and Choi and Huh (2017).

Trades through Separate Agent

- For trades that go through dealers that act purely as an agent, the total customer trading costs consist of two spreads:
  - Agent-dealer spread:
    - Customer-interfacing dealer, A or C, acting as an agent.
  - Principal-dealer spread:
    - Liquidity providing dealer, B, who holds bonds in inventory and acts purely as a market-maker.
  - The total transaction costs are sum of the two, rather than each separately an independent single transaction costs, as is estimated in extant literature.

Other Trades: Dual Capacity Dealers

- The majority of bond trades – 61% (66%) of customer buys (sells) in our sample period – do <u>not</u> go through a separate agent-dealer.
- They go through dual-capacity dealers - a dealer that sells directly to a customer from its bond inventory or buys bonds from a customer and retains the bonds in its inventory.
- In these cases, the dealer acts in dual capacity – as an agent for the customer and as a market making principal.

Agent and Principal Roles

- The agent and principal roles may differ from bond to bond and from trade to trade.
  - A dealer may be a principal-dealer for one set of bonds or trades, an agent-dealer for another set of bonds or trades, and a dual-capacity-dealer for a third set of bonds or trades.
  - For instance, a market-making dealer who trades with both agent-dealers and non-dealer customers functions as both a principal dealer and a dual-capacity-dealer.

Customer Interface Costs: Agent-dealers vs. Dual-capacity-dealers

- ► Net customer interface costs of an agent-dealer are likely to be different from those of a dual-capacity-dealer.
  - – Direct costs of managing customers should be similar.
  - – However, a customer interface can provide significant information benefits to a dealer acting in dual-capacity for the bond at that time.
    - Enable judgment about whether, and to what extent, trader is informed; and adjust spread accordingly.
    - Observe who is trading.
  - – Principal dealer does not have access to this information.
  - – Agent-dealer does not arguably care for the information since it does not have an inventory in the bond at that time.

Customer Interface Costs: Agent-dealers vs. Dual-capacity-dealers

- A principal-dealer's spread impounds the typical market-making costs – inventory risks and adverse selection costs.
- A dual-capacity-dealer's spread impounds:
  - Market-making costs
  - Customer-management benefits and costs.
- Difference between dual-capacity-dealer spreads and principal-dealer spreads proxies for net customer-interface costs (as relevant to dual-capacity-dealers.)

Different Dealer Spreads

- In this paper, we estimate agent-dealer, principal-dealer, and dual-capacity-dealer spreads, and analyze how they vary with the factors that are (or are not) relevant to the specific economic function (i.e., agent, or market-maker, or both) being fulfilled by them.
- Since the three roles are quite different, we expect the determinants of their respective spreads to differ: most specifically the impact of:
  - bond market risk factors relevant more for Principal-dealers and Dual-capacity dealers
  - bond liquidity measures relevant more for agent-dealers.

Table 2 - Descriptive Statistics for Paired and Unpaired Trades – Nov 2008 to Dec 2010

			Trade size in bonds		
Trade type	# trades	% of trades	Median	Mean	Std. dev.
Dual-capacity-dealer sale	1,344,597	23.03%	40	498.1	1085.2
Agent-dealer sale	871,349	14.92%	15	42.2	179.2
Principal-dealer sale	855,873	14.66%	16	42.9	181.4
Interdealer unpaired	926,404	15.86%	90	538.0	1048.7
Principal-dealer purchase	458,737	7.86%	11	71.4	329.6
Agent-dealer purchase	472,229	8.09%	10	69.4	324.3
Dual-capacity-dealer purchase	910,291	15.59%	100	681.3	1261.0
All	5,839,480	100.00%	25	330.0	887.7

Table 3 - Agent-dealer and principal-dealer spread statistics on trades involving both dealer types

	Agent-dealer spreads		Roundtrip principal-dealer spreads (same day trades)
	Sales	Purchases	
Mean spread	$0.743	$0.299	$1.195
Median	$0.600	$0.000	$0.895
Standard deviation	$0.743	$0.517	$1.302
% Zero	22.86%	51.47%	
observations	871,349	472,229	530,214

Agent-dealer spreads are sizable and comparable in magnitude to principal-dealer spreads, implying that dealers' agent-related costs, i.e., search and customer interface costs, are comparable to dealers' market-making costs.

Results: Descriptive Statistics

- Average agent-dealer spreads on riskless principal sales and purchases are $0.743 and $0.299 respectively.
- Zero spreads are observed on 22.9% of agent-dealer sales and 51.5% of agent-dealer purchases. The agent-dealer is likely providing this service as part of a bundle of services to its client compensated through wrap or other fees.
- Mean agent dealer spreads are $0.963 for sales and $0.616 for purchases with zero-spread observations excluded.
- Mean principal-dealer spread on same day trades is $1.20, as compared average round trip agent-dealer spreads of $1.58 (excluding zero agent-dealer spreads).

Impact of Market Risk Factors

- Market-making component of dealer costs represents dealer compensation for inventory and adverse selection costs.
  - Since principal-dealers and dual-capacity-dealers make markets and agent-dealers do not, we expect principal-dealer and dual-capacity-dealer spreads, but not agent-dealer spreads, to be positively related to measures of market price risks -- specifically VIX and the MOVE index of expected interest rate volatility.
  - Anticipating that price risks are higher on lower rated bonds and longer duration bonds, we also expect principal-dealer and dual-capacity-dealer spreads to be higher on lower rated and longer maturity bonds.

Impact of Bond Liquidity

- On the basis of theoretical search cost models, we expect search costs, and hence agent-dealer spreads, to be higher for assets with low liquidity -- specifically bonds with low ratings and longer terms-to-maturity.

Results: Agent- and Principal Dealer Spreads Impact of Risk

- Our hypothesis that due to inventory and asymmetric information risks, principal-dealer spreads should vary directly with bond market risk is strongly supported.
  - The coefficients of both the VIX and MOVE indices are large, positive, and significant at the .001 level.
    - A one standard deviation rise in the VIX (MOVE) raises the spread about $0.345 ($0.242).
    - Spreads were also higher in the financial crisis period.

Results: Agent- and Principal Dealer Spreads Impact of Risk

- We also find, consistent with our hypothesis, that since they face no inventory or information risk, agent dealer spreads are much less sensitive to the risk variables.
  - In regressions without time dummies, coefficients of VIX and MOVE variables are either insignificant or negative.
  - In regressions with time dummies, VIX's coefficients are positive and significant but MOVE's are negative and significant and VIX's coefficients are much smaller.
  - E,g., a one standard deviation change in VIX is associated with only a $0.026 change in the agent-dealer spread.

Results: Agent- and Principal Dealer Spreads Impact of Trading Volume

- Our hypothesis that, due to inventory costs, principal-dealer spreads vary inversely with how actively a bond is traded is also strongly supported.
  - A one standard deviation increase in trade volume tends to lower the principal-dealer spread about $0.382.
- Our hypothesis that agent-dealer spreads vary inversely with trading activity due to higher search costs on rarely traded bonds is also confirmed but we find that agent-dealer spreads are much less sensitive to trade volume than principal dealer spreads.
  - A one standard deviation increase in trade volume tends to lower agent-dealer spreads about $0.028.

Results: Agent- and Principal Dealer Spreads Impact of Trading Volume

- Several previous studies, e.g., Hong and Warga (2000), and Bessembinder, Maxwell, Venkataraman (2006), Henderschott et al. (2015), and Harris (2015) find that dealer spreads vary inversely with trading volume while Goldstein, Hotchkiss, and Sirri (2007) find a direct relationship. However, none heretofore have separated agent-dealer and principal-dealer spreads.

Results: Agent- and Principal Dealer Spreads Impact of Volatility

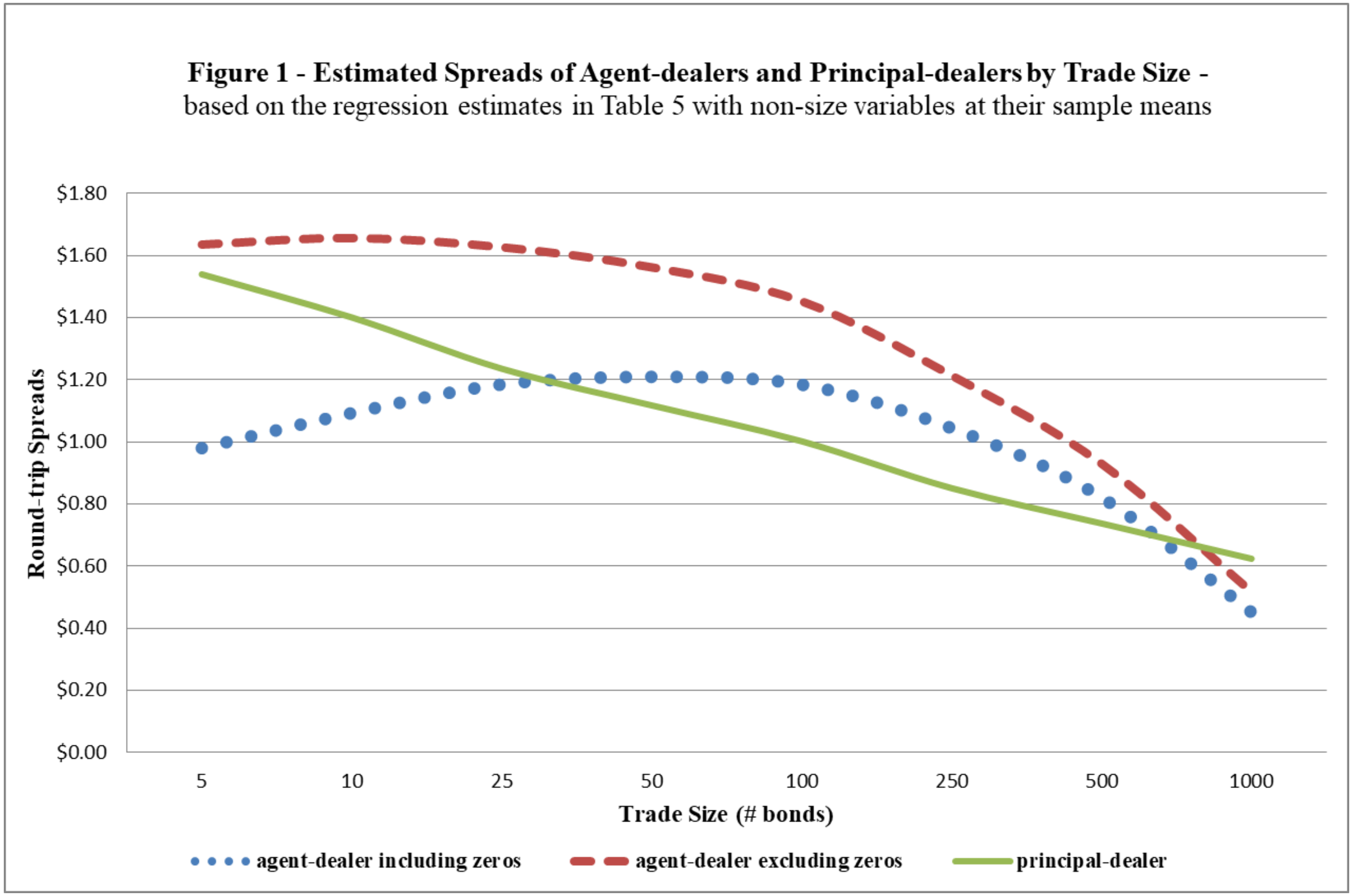
- Since price variability is higher on longer term and lower rated issues, principal-dealers' inventory risks are higher which should lead to higher spreads on these issues.
- Higher price variability likely raises the expected marginal payout to additional search leading to higher search costs for agent-dealers on longer term and lower rated issues.
- Our results are consistent with both hypotheses.
- Principal-dealer spreads are about $0.23 higher on 20-year bonds than on 5-year notes, and agent-dealer spreads (including zeros) are about $0.31 higher.

Results: Agent- and Principal Dealer Spreads Electronically Executed Trades

- Trading costs for agent-dealers should be lower for automated electronically-executed trades because of lower search costs.
- Consistent with this, we find that agent-dealer spreads are about \$0.21 to \$0.28 lower on automated electronically- executed trades.

Results: Agent- and Principal Dealer Spreads Electronically Executed Trades

- Interestingly, the cost saving on agent-dealer spreads for automated electronically-executed trades is partially offset by higher ($0.10 to $0.11) principal-dealer spreads.
- This suggests that principal-dealers may not post their best quotes in systems where free-standing quotes can be automatically executed, or when responses to requests-for-quote are instantly generated algorithmically.
  - When such principal dealers provide options to the market through their free-standing quotes or quotes that have been pre-programmed in advance, these options are long-horizon options.
  - When they revert with a quote in response to a specific request-for-quote in a non-automated system, the quotes have a short shelf-life.



Agent-dealers devote less search effort to smaller trades, and/or, larger traders have greater bargaining power.

Results: Dual-capacity Spreads

- As hypothesized, most of the results for dual-capacity-dealer trades are similar to those for principal-dealer trades.
- Again, spreads are positively related to the VIX and MOVE indices, and negatively related to trading volume although the relations are somewhat weaker than observed for principal-dealer spreads.
- Similarly, spreads are higher in 2008 and early 2009 than later.
- As with principal-dealers, dual-capacity-dealer spreads are higher on lower rated and longer maturity bonds and decline as trade size increases.

Customer Interface Costs and Benefits

- By interacting directly with the trader, the dual-capacity dealer may potentially know the identity of the trader, judge the extent to which the trader is informed, and thereby (at least) partially infer the private information in the trade.
  - In trades through an agent-dealer, the associated principal-dealer does likely not get access to this information.
- We accordingly expect that the dual-capacity-dealer's customer interface benefit to be greater (and net customer interface costs to be lower) for bonds with high information asymmetry, e.g., lower rated bonds or bonds that are rarely traded, and also when market uncertainty as represented by the VIX and MOVE indices is high.

Results: Customer Interface Costs

- Consistent with information asymmetry being higher on low rated and rarely traded bonds, we find that net customer interface costs of dual-capacity-dealers increase with trading volume and decline as the credit rating worsens.
- Consistent with the benefits of knowing the customer and how informed s/he is are greater when risk and information asymmetry are high, we find net interface costs to be negatively correlated with the VIX and MOVE indices.
- A one standard deviation increase in MOVE or VIX decreases customer net interface costs (and hence arguably increases information benefits for dual-capacity-dealers from customer interface) by \$0.212 and \$0.083 respectively.

Results: Customer Interface Costs

- Net customer interface costs are also significantly lower for bonds with greater credit risk.
- Net customer interface costs are relatively large in magnitude for small trades and decrease sharply on a per-bond basis as trade size increases.
- This is consistent both with customer interface costs having a fixed per trade component and/or with dual-capacity-dealers deriving more informational benefits from knowing the trader’s identity on larger trades.

Independent Variables	Agent-dealer Spreads	Principal-dealer Spreads
Risk		
VIX and MOVE indices		+
Dummies for 2008 Financial Crisis		+
Liquidity -		
Log of average daily trading volume - 6 months	-	-
Liquidity and Information Asymmetry		
Log of years to maturity	+	+
Rating (AAA=1, AA+=2, etc.)	+	+
Trade size (3 variables)	both negative but non-linear	
Average change in price of bonds with same rating and approximate maturity between trade dates	NA	+

Transaction Cost Estimates

- Finally, we find that most earlier estimations of bond market transaction costs from TRACE data tend to significantly underestimate total transaction costs because they failed to recognize that customer transactions often involve two component trades and spreads.
- This helps explain the wide variation in previous bond transaction cost estimates.

Table 8 - Transaction Cost Relations based on Successive Trades over the Complete Sample

	Principal-dealer spreads	Agent-dealer spreads	Dual-capacity dealer spreads	Implied customer interface costs of dual-capacity-dealers
Intercepts (• $_{m}$)	0.3015** (.0152)	-0.0649** (.0155)	2.5311** (.0131)	2.2296** (.0182)
VIX index (x100)	2.0707** (.0222)	-0.0342 (.0206)	1.3507** (.0185)	-0.7200** (.0256)
MOVE index (x100)	0.7286** (.0072)	0.0017 (.0067)	0.1199** (.0057)	-0.6087** (.0083)
log of avg. daily bonds traded - 6 months	-0.1599** (.0010)	-0.0380** (.0012)	-0.0635** (.0012)	0.0973** (.0013)
log of years to maturity	0.2976** (.0028)	0.5364** (.0026)	0.4370** (.0021)	0.1393** (.0032)
average rating	0.0435** (.0005)	0.0287** (.0005)	0.0363** (.0003)	-0.0072** (.0006)
Trade size variables not shown				
Average • P for same rating/maturity bonds	0.4382** (.0041)			
Adjusted r-square	.437			

Implications

- The extensive extant empirical research on dealer spreads in OTC markets need reinterpretation in the context of the large agent-related costs that we find, costs that meaningfully vary with several relevant economic factors in a manner that is similar to market-making costs in some cases, and differently in many other cases.
- Second, our results on the possible informational benefits to dual-capacity-dealers of directly interacting with customers – should also be applicable to other OTC markets.

Implications

- Third, with traditional OTC voice-based request-for-quote markets being increasingly substituted by automated electronic execution methods, our result that principal-dealers may not post their best quotes in systems where standing quotes can be automatically executed, may again have wider applicability to other OTC markets.

Conclusions

- Our first major contribution is to estimate and test hypotheses on agent-dealer spreads.
  - Agent-dealer spreads are sizable and comparable in magnitude to principal-dealer spreads.
  - Since they do not bear any inventory risks, agent-dealer spreads are not related with measures of market uncertainty.
  - Agent-dealer spreads are lower for trades that provide a direct proxy for lower search and order processing costs: i.e., automated electronic trades.
  - There is supporting evidence on the impact of search and order-processing costs through proxies for liquidity: agent-dealer spreads increase as bond specific trading volume decreases, and for lower rated and longer maturity bonds.

Conclusions

- Our second contribution is to cleanly test hypotheses on market-making.
  - Both principal-dealer spreads and dual-capacity dealer spreads are, as expected, significantly positively related with the VIX and the MOVE indices of volatility and market uncertainty.
  - They are also negatively related to bond specific trading volume and are higher for lower rated and longer maturity bonds for both liquidity and risk reasons.
  - But are higher for automated electronically-executed trades, suggesting that principal-dealers may not post their tightest quotes in automated systems where they are sitting ducks.

Conclusions

- Our third contribution is to investigate the dual-capacity feature of dual-capacity-dealers.
  - Dual-capacity-dealers extract significant information-related benefits from their direct interaction with trading customers.
    - Our evidence indicates that this benefit is greatest on large trades, when market uncertainty is high, and on issues with more information asymmetry.
  - Typically, customers face significantly lower total explicit trading costs if they trade directly with a dual-capacity-dealer rather than through an agent-dealer.
  - Similar to principal-dealers, dual-capacity-dealer spreads are related positively with VIX and MOVE volatility indices, negatively related to bond specific trading volume, and are higher for lower rated and longer maturity bonds.

Conclusions

- Finally, we find that most earlier estimations of bond market transaction costs from TRACE data tend to significantly underestimate total transaction costs because they failed to recognize that customer transactions often involve two component trades and spreads.

- Our results have important implications for the extensive extant empirical research on dealer spreads in other OTC markets since these studies have also not separated dealers' agent-related and market-making roles and costs and have tended to focus solely on dealers' making-making role.